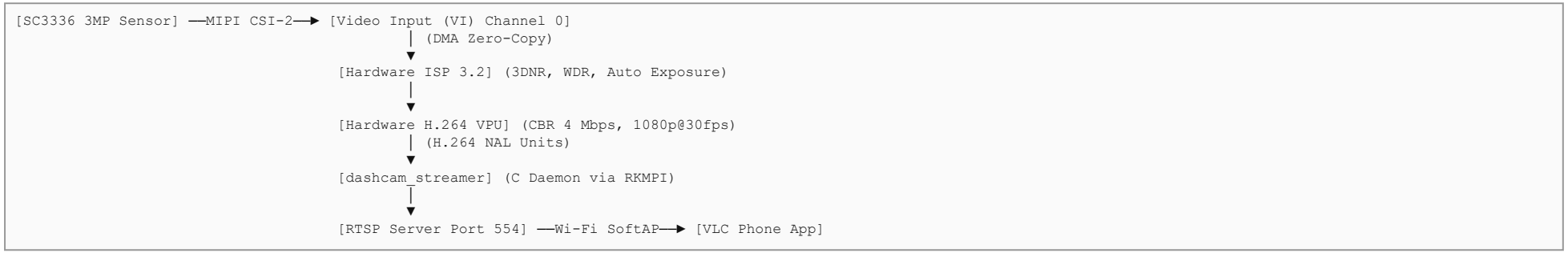
EMBEDDED SOFTWARE ARCHITECTURE & VIDEO PIPELINE (SHEET 1)

1. OPERATING SYSTEM & FLASH MEMORY PARTITIONING

Firmware executes from 128MB SPI NAND Flash under a lean Buildroot Linux distribution optimized for sub-6-second cold boot times:

PARTITION	SIZE	FILESYSTEM / IMAGE	FUNCTIONAL CONTENTS
mtd0 (U-Boot)	4 MB	Raw Binary	Rockchip Second-Stage Bootloader & DDR initialization.
mtd1 (Kernel)	12 MB	Flattened Device Tree + zImage	Linux 5.10 Kernel + Rockchip Media Driver & RTL8188 Wi-Fi Driver.
mtd2 (Rootfs)	64 MB	UBIFS (Read-Only)	Busybox system utilities, RKMPI shared libraries, hostapd, udhcpd.
mtd3 (Userdata)	48 MB	UBIFS (Read-Write)	Network credentials, streamer configuration, persistent logs.

2. HARDWARE VIDEO COMPRESSION ARCHITECTURE



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DOCUMENT TITLE **EMBEDDED SOFTWARE ARCHITECTURE**	REV **1.4**	DESIGN ENGINEER **Souvik H**	ENGINEER REGISTRATION NO. **IN-ENG-2022-22824**		
		ENGINEERING DISCIPLINE **EMBEDDED SOFTWARE & FIRMWARE**	STATUS **RELEASED**		

EMBEDDED SOFTWARE ARCHITECTURE & VIDEO PIPELINE (SHEET 2)

3. CORE VIDEO STREAMER IMPLEMENTATION LOGIC (`DASHCAM_STREAMER.C`)

```
// RKMPI Zero-Copy Hardware Pipeline Setup
RK_MPI_SYS_Init();

// Configure Video Input (VI) Channel 0
VI_CHN_ATTR_S vi_attr;
vi_attr.stIspOpt.u32BufCount = 2; // Strict double buffering for 64MB DRAM
vi_attr.stSize.u32Width = 1920;  vi_attr.stSize.u32Height = 1080;
vi_attr.enPixFmt = IMAGE_TYPE_NV12;
RK_MPI_VI_SetChnAttr(0, 0, &vi_attr);
RK_MPI_VI_EnableChn(0, 0);

// Configure Hardware H.264 VPU Video Encoder
VENC_CHN_ATTR_S venc_attr;
venc_attr.stVencAttr.enType = RK_VIDEO_ID_AVC;
venc_attr.stVencAttr.u32Profile = 77; // H.264 Main Profile
venc_attr.stVencAttr.u32PicWidth = 1920; venc_attr.stVencAttr.u32PicHeight = 1080;
venc_attr.stRcAttr.enRcMode = VENC_RC_MODE_H264CBR;
venc_attr.stRcAttr.stH264Cbr.u32BitRate = 4000; // 4 Mbps
venc_attr.stRcAttr.stH264Cbr.u32Gop = 30;       // 1 Keyframe per sec
RK_MPI_VENC_CreateChn(0, &venc_attr);

// Hardware Bind: VI Channel 0 -> VENC Channel 0
MPP_CHN_S src = { .enModId = RK_ID_VI,   .s32DevId = 0, .s32ChnId = 0 };
MPP_CHN_S dst = { .enModId = RK_ID_VENC, .s32DevId = 0, .s32ChnId = 0 };
RK_MPI_SYS_Bind(&src, &dst);
```

4. BOOT AUTOMATION SCRIPT (`/ETC/INIT.D/S99DASHCAM`)

Initializes wireless interface into SoftAP mode and launches RTSP streaming service at boot:

```
modprobe 8188eu 2>/dev/null
ifconfig wlan0 192.168.169.1 netmask 255.255.255.0 up
hostapd -B /etc/hostapd.conf
udhcpd /etc/udhcpd.conf
echo 48 > /sys/class/gpio/export && echo out > /sys/class/gpio/gpio48/direction && echo 1 > /sys/class/gpio/gpio48/value
/usr/bin/dashcam_streamer &
```

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